

Big Data and Automated Content Analysis (6 ECTS)

Cursusdossier

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Chapter 1

Short description of the course

“Big data” refers to data that are more voluminous, but often also more unstructured and dynamic, than traditionally the case. In Communication Science and the Social Sciences more broadly, this in particular concerns research that draws on Internet-based data sources such as social media, large digital archives, and public comments to news and products. This emerging field of studies is also called *Computational Social Science* (Lazer et al., 2009) or, narrowed down to the analysis of communication, *Computational Communication Science* (Shah, Cappella, & Neuman, 2015).

The course will provide insights in the concepts, challenges and opportunities associated with data so large that traditional research methods (like manual coding) cannot be applied any more and traditional inferential statistics start to lose their meaning. Participants are introduced to strategies and techniques for capturing and analyzing digital data in communication contexts. We will focus on (a) data harvesting, storage, and preprocessing and (b) computer-aided content analysis, including natural language processing (NLP) and computational social science approaches. In particular, we will use advanced machine learning approaches and models like word embeddings.

To participate in this course, students are expected to be interested in learning how to write own programs in Python. Some basic understanding of programming languages is helpful, but not necessary to enter the course. Students without such knowledge are encouraged to follow one of the many (free) online introductions to Python to prepare.

Chapter 2

Exit qualifications

The course contributes to the following three exit qualifications of the Research Master in Communication Science:

Expertise in empirical research

3. Knowledge and Understanding: Have in-depth knowledge and a thorough understanding of advanced research designs and methods

4. Skills and abilities: Are able, independently and on their own, to set up, conduct, report and interpret advanced academic research.

Academic abilities and attitudes

6. Attitude: Accept that scientific knowledge is always 'work in progress' and that arguments must be considered and conclusions drawn on the basis of empirical results and valid criticism.

The exit qualifications are elaborated in the following 11 specifications:

3. Knowledge and Understanding: Have in-depth knowledge and a thorough understanding of advanced research designs and methods.

3.1. Have in-depth knowledge and a thorough understanding of advanced research designs and methods, including their value and limitations.

3.2. Have in-depth knowledge and a thorough understanding of advanced techniques for data analysis.

4. Skills and abilities: Are able, independently and on their own, to set up, conduct, report and interpret advanced academic research.

4.1 Are able to formulate research questions and hypotheses for advanced empirical studies

4.2 Are able to develop a research plan, choose appropriate and suitable research designs and methods for advanced empirical studies, and justify the underlying choices.

4.3 Are able to assess the validity and reliability of advanced empirical research, and to judge the scientific and professional value of findings from advanced empirical research.

4.4 Are able to apply advanced empirical research methods.

6. Academic attitudes

6.1 Regularly assesses their own assumptions, strengths and weaknesses critically.

6.2 Accepts that scientific knowledge is always 'work in progress' and that something regarded as 'true' may be proven to be false, and vice-versa.

6.3 Are keen to acquire new knowledge, skills and abilities.

6.4 Are willing to share and discuss arguments, results and conclusions, including submitting one's own work to peer review.

6.5 Are convinced that academic debates should not be conducted on the basis of rhetorical qualities but that arguments must be considered and conclusions drawn on the basis of empirical results and valid criticism.

Chapter 3

Testable objectives

3. Knowledge and Understanding: Have in-depth knowledge and a thorough understanding of advanced research designs and methods.

3.1. Have in-depth knowledge and a thorough understanding of advanced research designs and methods, including their value and limitations.

3.2. Have in-depth knowledge and a thorough understanding of advanced techniques for data analysis.

A Students can explain *fundamental research designs* and *commonly employed methods* in existing research articles on Big Data and automated content analysis.

B Students can on their own and in own words critically discuss the pros and cons of *fundamental* research designs and methods employed in existing research articles on Big Data and automated content analysis; they can, based on this, give a critical evaluation of the methods and, where relevant, give advice to improve the study in question.

C Students are able to identify some *basic* techniques from the field of computer science and computer linguistics that are applicable to research in communication science; they can explain the principle of some traditional approaches to text analysis, namely simple rule-based techniques and basic methods of unsupervised and supervised machine learning.

4. Skills and abilities: Are able, independently and on their own, to set up, conduct, report and interpret advanced academic research.

4.1 Are able to formulate research questions and hypotheses for advanced empirical studies

4.2 Are able to develop a research plan, choose appropriate and suitable research designs and methods for advanced empirical studies, and justify the underlying choices.

4.3 Are able to assess the validity and reliability of advanced empirical research, and to judge the scientific and professional value of findings from advanced empirical research.

4.4 Are able to apply advanced empirical research methods.

- D Students can on their own formulate a research question and hypotheses for own empirical research in the domain of Big Data.
 - E Students can on their own chose, execute and report on *fundamental* research methods in the domain of Big Data and automatic content analysis.
 - F Students know how to collect data with APIs or read in existing data files; they know how to analyze these data with fundamental automated techniques and to this end, they have basic knowledge of the programming language Python and know how to use fundamental Python-modules for communication science research.
6. Academic attitudes
- 6.1 Regularly asses their own assumptions, strengths and weaknesses critically.
 - 6.2 Accept that scientific knowledge is always 'work in progress' and that something regarded as 'true' may be proven to be false, and vice-versa.
 - 6.3 Are keen to acquire new knowledge, skills and abilities.
 - 6.4 Are willing to share and discuss arguments, results and conclusions, including submitting one's own work to peer review.
 - 6.5 Are convinced that academic debates should not be conducted on the basis of rhetorical qualities but that arguments must be considered and conclusions drawn on the basis of empirical results and valid criticism.
- G Students can critically discuss strong and weak points of their own research using fundamental techniques from the field of Big Data and Automated Content Analysis, and suggest improvements.
 - H Students participate actively: reading the literature carefully and on time, completing assignments carefully and on time, active participation in discussions, and giving feedback on the work of fellow students give evidence of this.

Chapter 4

Planning of testing and teaching

The seminar consists of 12 meetings, two per week. Each week, in the first meeting, the instructor will give short lectures on the key aspects of the week, followed by seminar-style discussions. Theoretical considerations regarding Big Data and Automated Content Analysis are discussed, and techniques for analyzing Big Data are presented. We also discuss examples from the literature, in which these techniques are applied.

The second meetings each week are practicum-meetings, in which the students will apply what the techniques they have learned to their own or provided data sets. Here, they can also deepen their understanding of software tools, prepare their individual projects and get hands-on help. While there are in-class assignments as well as occasional assignments for at home (e.g., completing an online-tutorial to prepare for class), these are not graded.

To complete the course, next to active participation, the students have to successfully complete two summative graded assignments: one mid-term take-home exam and an individual project, in which they derive an empirical question from a theoretical starting point, and then conduct an Automated Content Analysis to answer the question. See Chapter 7 for details.

Chapter 5

Literature

The following schedule gives an overview of the topics covered each week, the obligatory literature that has to be studied each week, and other tasks the students have to complete in preparation of the class. In particular, the schedule shows which chapter of van Atteveldt, Trilling, and Arcila Calderón (2022) will be dealt with. Note that some basic chapters that explain how to install the software we are going to use have to be read before the course starts.

Next to the obligatory literature, the following books provide the interested student with more and deeper information. They are intended for the advanced reader and might be useful for final individual projects, but are by no means required literature. Bear in mind, though, that you may encounter slightly outdated examples (e.g., Python 2, now-defunct APIs etc.).

- McKinney, 2012: A lot of examples for data analysis in Python. A HTML-based version of the book can be viewed for free at <https://wesmckinney.com/book/>.
- VanderPlas, 2016: A more recent book on numpy, pandas, scikit-learn and more. It can also be read online for free on <https://jakevdp.github.io/PythonDataScienceHandbook/>. The contents are available as Jupyter Notebooks as well, see <https://github.com/jakevdp/PythonDataScienceHandbook>.
- The pandas cookbook by Julia Evans, a collection of notebooks on github: <https://github.com/jvns/pandas-cookbook>.
- Hovy, 2020: A thin book on bottom-up text analysis in Python with both a bit more math background and ready-to-use Python code implementations.

- Salganik, 2017: Not a book on Python, but on research methods in the digital age. Very readable, and a lots of inspiration and background about techniques covered in our course.

Chapter 6

Specific course timetable

Before the course starts: Prepare your computer.

✓ CHAPTER 1: INTRODUCTION

Make sure that you have a working Python environment installed on your computer. You cannot start the course if you have not done so.

Week 1: Programming for Computational (Communication|Social) Scientists

Monday, 31–3. Lecture.

✓ KITCHIN (2014)

✓ HILBERT ET AL. (2019)

We discuss what Big Data and Computational (Social|Communication) Science are. We talk about challenges and opportunities as well as the implications for the social sciences in general and communication science in particular. We also pay attention to the tools used in CSS, in particular to the use of Python.

Additionally, the journal *Communication Methods and Measures* had a special issue (volume 12, issue 2–3) about Computational Communication Science. Read at least the editorial (van Atteveldt & Peng, 2018), but preferably, also some of the articles (you can also do that later in the course).

Towards the end of the lecture, we will make first contact with writing code.

Thursday, 3–4. Lecture with exercises.

- ✓ CHAPTER 3: PROGRAMMING CONCEPTS FOR DATA ANALYSIS
- ✓ CHAPTER 4: HOW TO WRITE CODE

You will get a very gentle introduction to computer programming. During the lecture, you are encouraged to follow the examples on your own laptop. We will do our first real steps in Python and do some exercises to get the feeling with writing code.

Week 2: From files and APIs to lists, dictionaries, or dataframes

- ✓ CHAPTER 5: FROM FILE TO DATAFRAME AND BACK
- ✓ FREELON (2018-10)

We talk about file formats such as `csv` and `json`; about encodings; about reading these formats into basic Python structures such as dictionaries and lists as opposed to reading them into dataframes; and about retrieving such data from local files, as parts of packages, and via an API.

Monday 7–4. Lecture.

A conceptual overview of different file formats and data sources, and some practical guidance on how to handle such data in basic Python and in Pandas.

Thursday 10–4. Lab session.

We will exercise with the data structures we learned in week 1, as well as with different file formats.

Week 3: Data wrangling and exploratory data analysis

Of course, you don't need Python to do statistics. Whether it's R, Stata, or SPSS – you probably already have a tool that you are comfortable with. But you also do not want to switch to a different environment just for getting a correlation. And you definitely don't want to do advanced data wrangling in SPSS... This week, we will discuss different ways of organizing your data

(e.g., long vs wide formats) as well as how to do conventional statistical tests and simple plots in Python.

Monday, 14–4. Lecture.

- ✓ CHAPTER 6: DATA WRANGLING
- ✓ CHAPTER 7.1. SIMPLE EXPLORATORY DATA ANALYSIS
- ✓ CHAPTER 7.2. VISUALIZING DATA

We will learn how to get your data in the right shape and how to get a first understanding of your data, using exploratory analysis and visualization techniques. We will cover data wrangling with pandas: converting between wide and long formats (melting and pivoting), aggregating data, joining datasets, and so on.

Thursday, 17–4. Lab session.

We will apply the techniques discussed during the lectures to multiple datasets.

Week 4: Processing textual data

In this week, we will dive into how to deal with textual data. How is text represented, how can we process it, and how can we extract useful information from it? Unfortunately, text as written by humans usually is pretty messy. We will therefore dive into ways to represent text in a clean(er) way. We will introduce the Bag-of-Words (BOW) representation and show multiple ways of transforming text into matrices.

Monday, 21–4. No Lecture: Easter Monday

Thursday, 24–4. Lecture

- ✓ CHAPTER 9: PROCESSING TEXT
- ✓ CHAPTER 10: TEXT AS DATA
- ✓ CHAPTER 11, SECTIONS 11.1–11.3: AUTOMATIC ANALYSIS OF TEXT

Additional recommended background reading on stopwords: Nothman, Qin, and Yurchak (2018).

This lecture will introduce you to techniques and concepts like lemmatization, stopword removal, n-grams, word counts and word co-occurrences, and

regular expressions. We then proceed to introducing BOW representations of text.

As an exercise, we will write a first automated content analysis script.

Take home exam

In week 4, the midterm take-home exam is distributed after the Thursday meeting (24–4). The answer sheets and all files have to be handed in no later than Sunday evening (27–4, 23.59).

Week 5: Education free week

Monday, 28–4. No Lecture

Thursday, 1–5. No lab session

Week 6: Unsupervised approaches to text analysis

In this week, we will make the transition from classic statistical modeling as you know it from your previous courses to machine learning. We will discuss how both approaches are related (or even identical) and where the differences are.

Monday, 5–5. No lecture: Liberation day

Thursday, 8–5. Shorter lecture combined with exercises.

✓ CHAPTER 11.5. UNSUPERVISED TEXT ANALYSIS: TOPIC MODELING AND BEYOND

✓ MAIER ET AL. (2018)

✓ KROON, WELBERS, TRILLING, AND VAN ATTEVELDT (2024) Read the section *Limitations of Traditional Content Analysis Techniques* (pp. 144–152, up to and including Table 1).

We will discuss the use of unsupervised models for the explorative analysis of text. A first approach that has historically been employed to do this is to simply apply unsupervised methods such as PCA and k-means clustering on a BOW representation of text – something that you could actually have done already with your knowledge from Part I. Starting from there, we proceed to

discuss a second approach, Latent Dirichlet Allocation (LDA), also referred to as (a form of) topic modeling. Both approaches have been influential for the field, but are less of a silver bullet than many students and researchers seem to think. We will therefore introduce a much more state-of-the-art approach that is build on top of a pre-trained Transformer instead of relying on a BOW representation.

As we do not have a lecture on Monday, we will provide a short introduction into unsupervised machine learning, what it is used for, and how they can be evaluated. In addition, we will provide some hands-on excersises to start experimenting with topic modelling.

Week 7: Supervised approaches to text analysis

During this week, we will discuss the basics of machine learning. You will be introduced to scikit-learn (Pedregosa et al., 2011), one of the most well-known machine learning libraries. We do not have the time to discuss machine learning techniques in depth. Rather, a practical and hands-on introduction is provided.

Monday, 12–5. Lecture

✓ CHAPTER 8: STATISTICAL MODELING AND SUPERVISED MACHINE LEARNING

✗ (YOU CAN SKIP 8.4 DEEP LEARNING)

✓ BOUMANS AND TRILLING (2016)

We will discuss the basics of supervised machine learning, and how its performance can be evaluated.

Thursday, 15–5. Lab session

Exercises with scikit-learn.

Week 8: Wrapping up

Monday, 19–5. Presentations of Final Projects

During this meeting, everybody will shortly pitch their final projects. This way, you will be able to recieve some feedback on your palms.

Final project

Deadline for handing in: Wednesday, 28-5, 23.59.

Chapter 7

Testing

An overview of the testing is given in the Test Matrix, displayed in Table 7.1.

Grading

The final grade of this course will be composed of the grade of one mid-term take home exam (30%) and one individual project (70%).

Mid-term take-home exam (30%)

In the mid-term take-home exam, students will show their understanding of the literature and prove they have gained new insights during the lecture/seminar meetings. They will be asked to critically assess various approaches to Big Data analysis and make own suggestions for research. Additionally, they need to (partly) write the code to accomplish this.

Grading criteria are communicated to the students together with the assignment, but in general are: For literature-related tasks in the exam:

- usage of specific examples from the literature;
- critique of different approaches;
- nameing of pro's, con's, potential pitfalls, and alternatives;
- giving practical advice and guidance.

For programming-related tasks in the exam:

- correctness, efficiency, and style of the code

Table 7.1: Test Matrix

	In-class assignments, reviewing work of fellow students, active participation	Mid-term take home exam	Final individual project
	<i>precondition</i>	<i>30% of final grade</i>	<i>70% of final grade</i>
A. Students can explain fundamental research designs and commonly employed methods in existing research articles on Big Data and automated content analysis	X	X	
B. Students can on their own and in own words critically discuss the pros and cons of fundamental research designs and methods employed in existing research articles on Big Data and automated content analysis; they can, based on this, give a critical evaluation of the methods and, where relevant, give advice to improve the study in question.	X	X	
C. Students are able to identify some basic techniques from the field of computer science and computer linguistics that are applicable to research in communication science; they can explain the principle of some traditional approaches to text analysis, namely simple rule-based techniques and basic methods of unsupervised and supervised machine learning.	X	X	X
D. Students can on their own formulate a research question and hypotheses for own empirical research in the domain of Big Data.			X
E. Students can on their own chose, execute and report on fundamental research methods in the domain of Big Data and automatic content analysis.			X
F. Students know how to collect data with APIs or read in existing data files; they know how to analyze these data with fundamental automated techniques and to this end, they have basic knowledge of the programming language Python and know how to use fundamental Python-modules for communication science research.			X
G. Students can critically discuss strong and weak points of their own research using fundamental techniques from the field of Big Data and Automated Content Analysis, and suggest improvements.			X
H. Students participate actively: reading the literature carefully and on time, completing assignments carefully and on time, active participation in discussions, and giving feedback on the work of fellow students give evidence of this.	X		

- correctness, completeness, and usefulness of analyses applied.

For conceptual and planning-related tasks:

- feasibility
- level of specificity
- explanation and argumentation why a specific approach is chosen
- creativity.

Final individual project (70%)

The final individual project typically consists of the following elements, which all contribute to the final grade:

- introduction including references to relevant (course) literature, an overarching research question plus subquestions and/or hypotheses (1–2 pages);
- an overview of the analytic strategy, referring to relevant methods learned in this course;
- carefully collected and relevant dataset of non-trivial size; here, using APIs, or combining existing datasets.
- a set of scripts for collecting, preprocessing, and analyzing the data using fundamental techniques discussed in the course. The scripts should be well-documented and tailored to the specific needs of the own project;
- output files;
- a well-substantiated conclusion with an answer to the RQ and directions for future research.

Depending on the chosen topic, the student will have to apply some of the techniques covered in the course. The assignment needs to present an explorative description of the dataset, combined with some of the fundamental techniques discussed during the course. During the lab sessions, as well as individual consultation, students may discuss the scope of the projects with the lecturer, the requirements that the specific project suggested by the student needs to fulfill, and the extent to which the different methods that the student plans to use will contribute to the final grade.

Grading and 2nd try

Students have to get a pass (5.5 or higher) for both mid-term take-home exam and the individual project. If the grade of one of these is lower, an improved version can be handed in within one week after the grade is communicated to the student. If the improved version still is graded lower than 5.5, the course cannot be completed. Improved versions of the final individual project cannot be graded higher than 6.0.

Chapter 8

Lecturers' team, including division of responsibilities

dr. Anne Kroon (teaching first half of the course)

dr. Felicia Loecherbach (teaching second half of the course) both lecturers
are involved in grading the assignments

Chapter 9

Calculation of students' study load (in hours)

- Elective total: 6 ECTS = 168 hours
- Reading:
 - 10 chapters, 5 articles, average 18.5 pages: around 280 pages. 6 pages per hour, thus 46.6 hours for the literature
 - Reading and doing tutorials: 40 hours for reading tutorials to acquire skills.
 - Reading/preparation total: 86.6 hours.
- Presence:
12*2 hours: 24 hours.
- Mid-term take-home exam, including preparation: 18 hours
- Final individual project, including data collection, analysis, write up:
39.4 hours

Total: 168 hours

Chapter 10

Calculation of lecturers' teaching load (in hours)

- Presence: 24 hours ($= 12 * 2$ hours)
- Preparation of weekly lectures, $7 * 4$ hours: 28 hours
- Preparation of weekly tutorials, $7 * 4$ hours: 28 hours
- Assisting students with setting up Python installation, individual help: 40 hours
- Feedback and grading take-home exams: $25 * 60$ minutes: 25 hours
- Feedback and grading final projects, including feedback on proposal and individual counseling: $25 * 60$ min: 25 hours
- Administration, e-mails, individual appointments: 20 hours

Total: 190 hours

Chapter 11

History of the course

In response to the feedback by the test review committee in 2019, the following changes were applied:

- Emphasized in the section “Testable Objectives” that – in contrast to the 12 ECTS course – students should be able to demonstrate, reflect, and apply knowledge about *fundamental* –rather than *advanced* techniques of automated content analysis. Likewise, testing will be based on the extent that students demonstrate knowledge and skills on the level of *fundamental* techniques. This sets the 6ec course apart from Big Data and Automated Content Analysis Part I and Part II, where students take a deeper dive into more advanced and recent techniques for automated content analysis.

In 2022, the course received a thorough update due to the publication of the new textbook (van Atteveldt et al., 2022).

In 2023, the course was slightly updated to incorporate recent developments in machine learning. Most elements remain the same—as the 6ECTS course focuses on fundamental techniques of automated content analysis. Yet, in week 7, we will discuss state of the art techniques for topic modelling. In addition, an outlook is provided on techniques that are the current becoming dominant in the field of machine learning.

In 2024, the course was slightly updated to make the course more AI proof (using hours from a TLC grant). The two main changes are: 1) The take-home exam will be more focused on debugging and explaining existing code over writing code and the testing will include a short video-walkthrough in addition to the written deliverable. This is aimed at ensuring some level of active engagement with the content by the students without having to use up additional in-class time. 2) One of the last sessions is being used to have students do a small pitch about their final project in class. This has two

reasons: First, the students see what their classmates are working on and can exchange ideas and second they can give and receive feedback on the final projects with enough time to still make changes and adjustments.

Literature

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